

Appendix A3: literature data treatment.

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Economic data

Economic information was derived from literature and from interviews. Values from literature were preferred in the case of reference values, such as mean annual expenses and fuel costs. Reference values were obtained from the interview only when no information was available in literature. Indications for expenses linked to extreme weather events were obtained from the interviews. Sources used were AER and Friskeverket.

Methods

The cost node reflects the economic impact of one single extreme weather event compared to annual average expenses. Cost categories to be used in the conditional probability table are negligible ($\text{cost} < 0.05\%$); low ($0.05\% < \text{cost} < 2.5\%$); medium ($2.5\% < \text{cost} < 5\%$); high ($> 5\%$).

The conditional probability tables are estimated through the function `cost.cpt`, here below reported. The input needed are:

- expenses (mu and sd): annual total expenses, including maintenance, fuel, personnel. This value is anchored in literature, when available;
- fuel.cost (mu and sd): average cost for single fishing trip. This value is anchored in literature, when available;
- damage.small (lo and hi): self declared cost for repairing a minor damage;
- damage.large (lo and hi): self declared cost for repairing a large damage;
- damage.unc: self declared uncertainty on the cost for repairing damages;
- fuel.avg (lo and hi): self declared fuel consumption for single average fishing trip;
- fuel.large (lo and hi): self declared fuel consumption for single longer-than-average fishing trip;

- economic.unc: self declared uncertainty associated with answer regarding expenses;
- fishing style: label describing the fishing style.

```
cost.cpt=function(expenses.mu, expenses.sd,
                  fuel.cost.mu,fuel.cost.sd,
                  damage.small.lo, damage.small.hi,damage.large.lo, damage.large.hi,
                  damage.unc,
                  fuel.avg.lo, fuel.avg.hi,
                  fuel.large.lo, fuel.large.hi,
                  miss.catch.mu,miss.catch.sd,
                  economic.unc,
                  f.style){

  # creating distributions
  ## distribution for annual expenses
  cost.annual=rtruncnorm(a=0, b=Inf,
                        mean=expenses.mu,
                        sd=expenses.sd, n=1000)

  ## distribution for minor damage
  cost.lo=rtruncnorm(a=damage.small.lo, b=damage.small.hi,
                    mean=(damage.small.hi-damage.small.lo)/2,
                    sd=damage.small.hi*damage.unc, n=1000)

  ## distribution for major damage
  cost.hi=rtruncnorm(a=damage.large.lo, b=damage.large.hi,
                    mean=(damage.large.hi-damage.large.lo)/2,
                    sd=damage.large.hi*damage.unc, n=1000)

  ## distribution for avg fuel
  fuel.lo=rtruncnorm(a=fuel.avg.lo, b=fuel.avg.hi,
                    mean=(fuel.avg.hi-fuel.avg.lo)/2,
                    sd=fuel.avg.hi*economic.unc, n=1000)

  ## distribution for large travel
  fuel.hi=rtruncnorm(a=fuel.large.lo, b=fuel.large.hi,
                    mean=(fuel.large.hi-fuel.large.lo)/2,
                    sd=fuel.large.hi*economic.unc, n=1000)

  # cost for fuel in extra travel
  extra.fuel.cost=(fuel.hi/fuel.lo)*rtruncnorm(a=0, b=Inf,
                                              mean=fuel.cost.mu,
                                              sd=fuel.cost.sd, n=1000)

  # missing catches
  miss.catch=rtruncnorm(a=0, b=Inf, mean=miss.catch.mu, sd=miss.catch.sd, n=1000)

  # create data frame
  cost.df=data.frame(minor.no=cost.lo, major.no=cost.hi, no.travel=extra.fuel.cost)
  cost.df$minor.travel=cost.df$minor.no+cost.df$no.travel
  cost.df$major.travel=cost.df$major.no+cost.df$no.travel
  cost.df.2=cost.df+miss.catch
  names(cost.df.2)=paste(names(cost.df.2), 'miss', sep='.')
```

```

cost.df.2$no.no.miss=miss.catch
cost.df=cbind(cost.df, cost.df.2)
cost.df[,]=apply(cost.df[,],2,function(x) x/cost.annual)
cost.df.disc=cost.df
cost.df.disc[,]=apply(cost.df.disc[,],2,
                      function(x) cut(x, c(-Inf, 0.005,0.025,0.05,Inf),
                                       labels=c('negligible','low','medium','high'))))

cost.df.disc$fishing_style=f.style
cost.cpt=cost.df.disc%>%
pivot_longer(-fishing_style, names_to = 'parent', values_to = 'state')%>%
dplyr::group_by(fishing_style,parent, state)%>%
tally()%>%
dplyr::mutate(prob=n/sum(n))

return(cost.cpt)
}

error.propagation=function(x.mu, x.er, y.mu, y.er){
  z.mu=x.mu/y.mu
  z.er=sqrt(((x.er/x.mu)^2)+((y.er/y.mu)^2))*z.mu
  return(data.frame(mu=z.mu, sd=z.er))
}

```

Coastal fisherman

Coastal fisherman economic information is explicitly available in the STECF_AER data source (STECF 2024). STECF_AER data are filtered for vessel length categories VL0010 and VL0812 and on fishing technologies FPO and DFN. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. Other values are provided in the interviews.

Table C.1: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
2	minor_damage_cost	10000	50000	sek_unit
3	minor_damage_cost	1	50000	sek_year
7	minor_damage_cost	1000	50000	sek_unit
2	major_damage_cost	100000	500000	sek_unit
3	major_damage_cost	50000	200000	sek
7	major_damage_cost	50000	100000	sek_unit
3	fuel_trip	1500	2500	sek_day
3	extra_fuel	3000	5000	sek_day
7	extra_fuel	50	100	B liter_trip

Table C.2: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	1706	1056	thousand SEK	STECF_AER
Fuel cost	4556	5020	SEK	STECF_AER

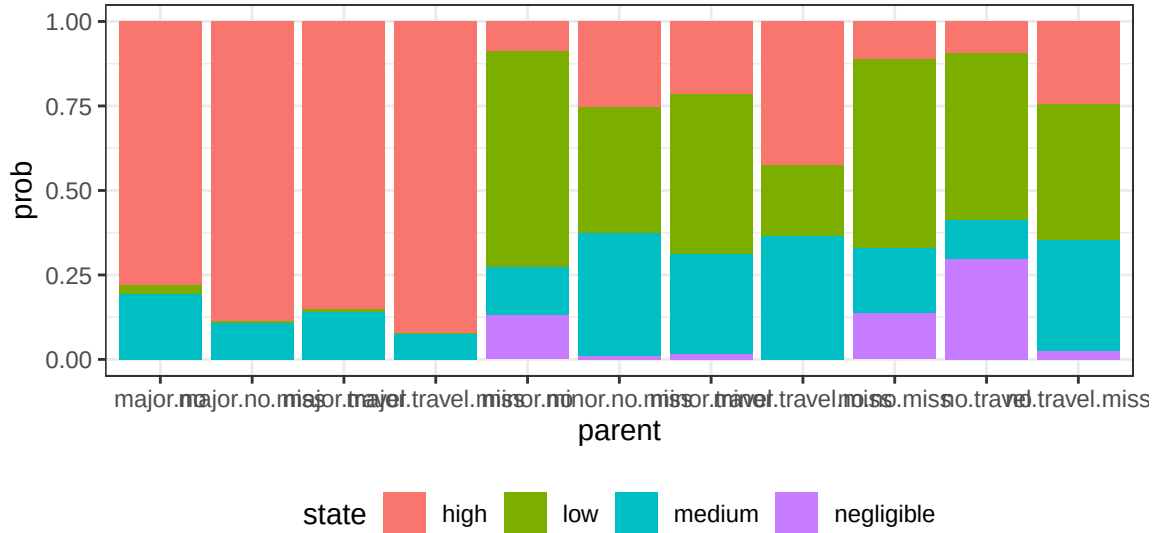


Figure C.1: CPT estimated for coastal fisherman

Pelagic trawlers

Pelagic economic information is derived from the STECF_AER data source (STECF 2024). STECF_AER data are filtered for vessel length categories VL2440, including all fishing technologies. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. The cost of longer travel was not available, and it is based on the assumption that an average travel costs between 80% and 120% of the average fuel cost, while a longer travel is between 100% and 150% of average fuel cost. Other values are provided in the interviews.

[1] 4980781 5284555 5844383 5749127 5516936

Table C.3: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
6	minor_damage_cost	10000	50000	sek_unit
6	major_damage_cost	100000	500000	sek_unit
6	fuel_trip	600	650	liters_hour
6	extra_fuel	NA	NA	not_reported

Table C.4: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	48993	5047	thousand SEK	STECF_AER
Fuel cost	208104	63543	SEK	STECF_AER

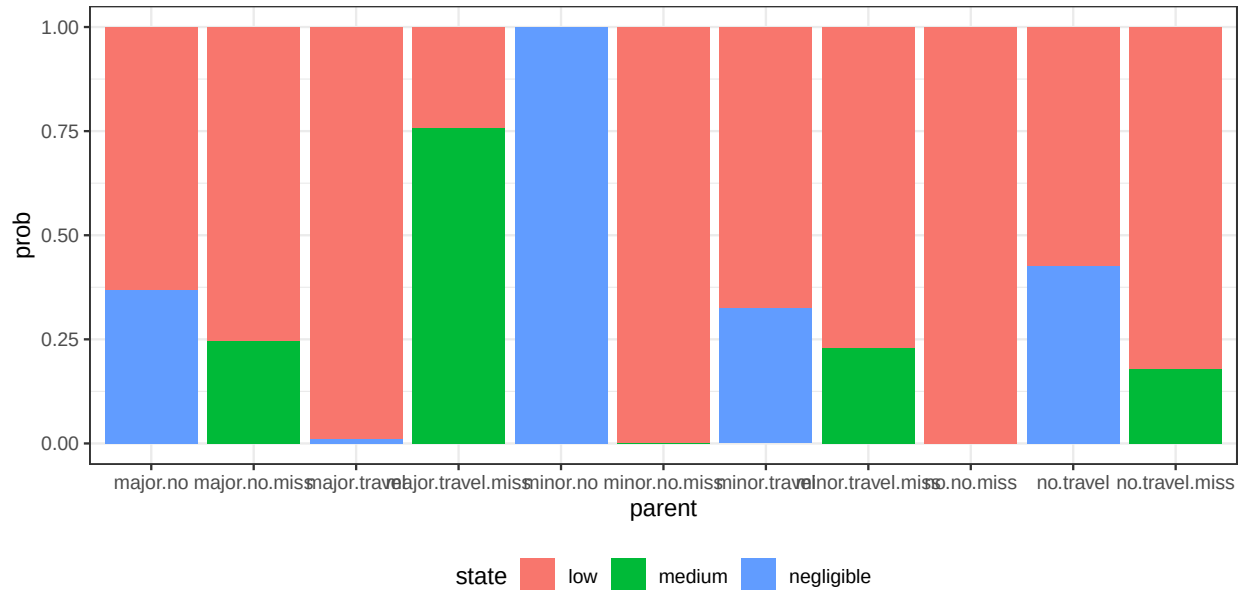


Figure C.2: CPT estimated for pelagic trawlers

Recreational

Pelagic economic information is derived from the recreational fisheries statistical report of the Swedish Department for Economic and Statistic Analysis (SWAM) data source (SWAM 2024). SWAM economic data represent the totality of recreational fisherman in Sweden, with no distinction between region or inland vs coastal fishing. Cost of fuel per trip is separately calculated for travel and for boat fuel. These two costs are then averaged. All the uncertainties provided in SWAM (on costs and on number of practitioners) are accounted for by propagating errors (REF). Annual spending is estimated by dividing total annual expenses by number of practitioners. Other values are provided in the interviews.

id_I	short_description	range.lwr	range.upr	unit_range
5	minor_damage_cost	10000	50000	sek_unit
5	major_damage_cost	50000	1000000	sek_unit
5	fuel_trip	30	50	B liter_trip
5	extra_fuel	20	50	B liter_trip

Table C.6: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	9213	3943	SEK	SWAM
Fuel cost	101	31	SEK	SWAM

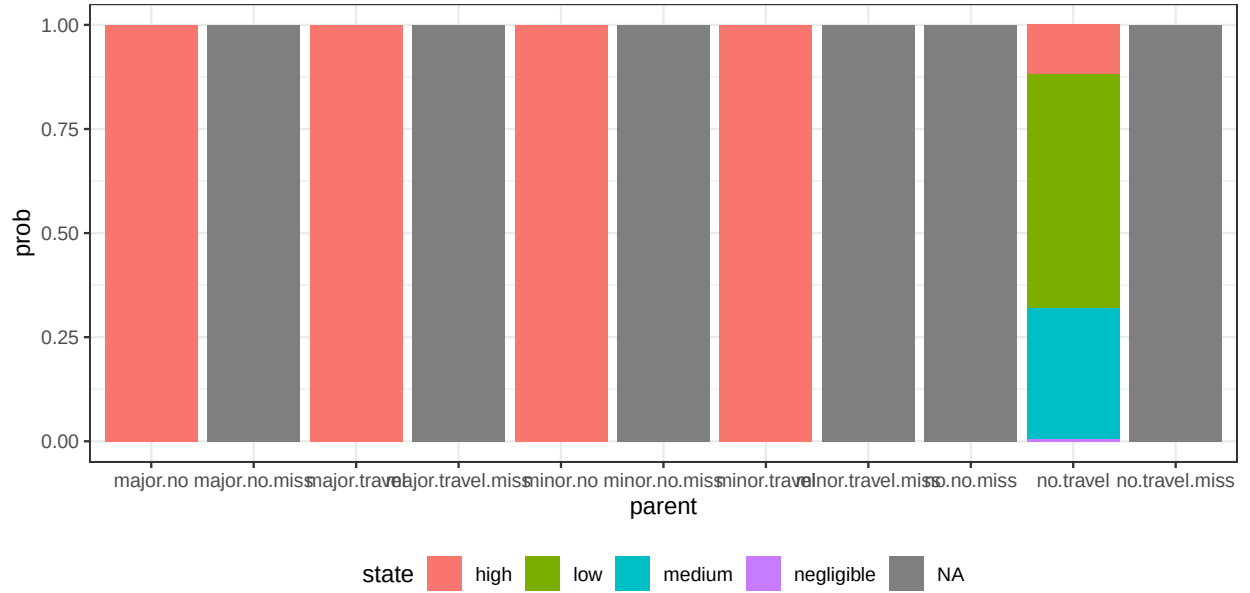


Figure C.3: CPT estimated for recreational fisherman

Fishing guide

No updated public information is available for fishing guide economic aspects in Sweden. All estimations base on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
1	season_maint	50000	10000	sek_season
1	minor_damage_cost	10000	20000	sek_unit
1	major_damage_cost	20000	6000000	sek_unit
1	fuel_trip	1000	1000	sek_day
1	extra_fuel	500	500	sek_day

Table C.8: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	174208	14889	SEK	Our estimation based on interviews

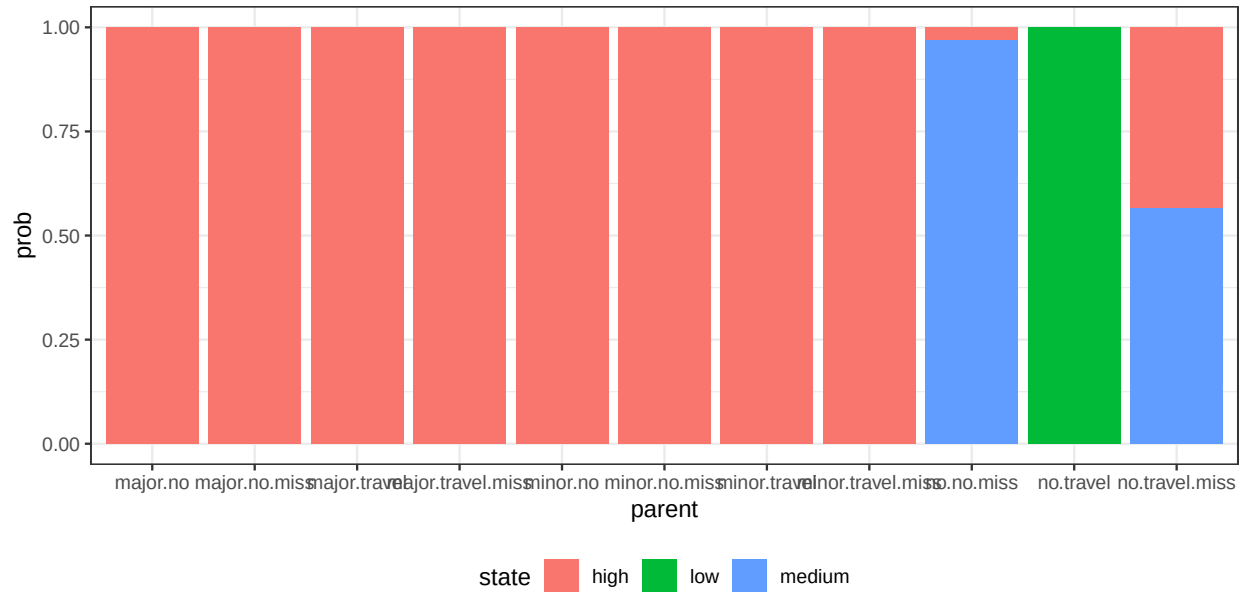


Figure C.4: CPT estimated for fishing guide

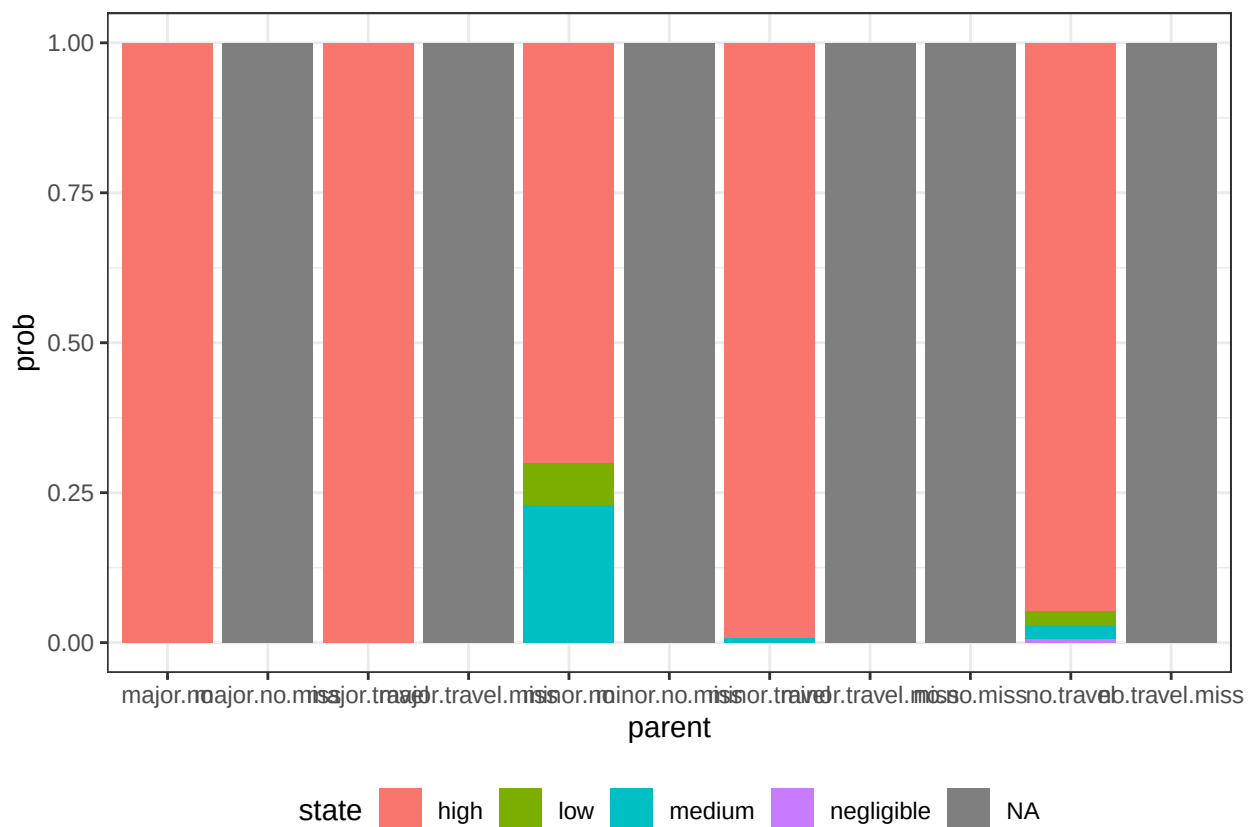
Archipelago

No updated public information is available for archipelago fishers economic aspects in Sweden. All estimations base on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
4	season_maint	NA	5000	sek_season
4	minor_damage_cost	200	2500	sek_unit
4	major_damage_cost	2500	5000	sek_unit
4	fuel_trip	5000	20000	sek_season
4	extra_fuel	NA	NA	not_reported

Table C.10: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	17277	4360	SEK	Our estimation based on interviews



STECF. 2024. *The 2024 Annual Economic Report on the EU Fishing Fleet (STECF 24-03 & 24-07)*. Luxembourg: Publications Office of the European Union. <https://publications.jrc.ec.europa.eu/repository/handle/JRC139642>.

SWAM. 2024. "Fritidsfiske 2024."